

JEE-Main-02-04-2026 (Memory Based)

[MORNING SHIFT]

Physics

Question: Dimensional formula for $\frac{1}{2} \in E^2$ is given as $MaLbTc$. Find the value of $2a-b+c$.

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (a)

Solution:

$$\begin{aligned}
 [\text{density}] &= \left[\frac{1}{2} \in E^2 \right] = \frac{ML^2T^{-2}}{L^3} \\
 &= ML^{-1}T^{-2} \\
 a &= 1, b = -1, c = -2 \\
 2a - b + c &= 2(1) - (-1) - 2 \\
 &= 1
 \end{aligned}$$

Question: The velocity of a particle is given as $\vec{v} = -x\hat{i} + 2y\hat{j} - 2z\hat{k} \text{ m/s}$.
The magnitude of acceleration at point (1,2,4) is _____ m/s^2 ?

Options:

- (a) $\sqrt{13}$
- (b) $\sqrt{65}$
- (c) 65
- (d) 13

Answer: (c)

Solution:

$$\vec{v} = x\hat{i} + 2y\hat{j} - 2\hat{k}$$

$$\vec{a} = x \frac{dx}{dx} \hat{i} + (2y) \frac{d(2y)}{dy} \hat{j} + 0$$

$$= x\hat{i} + (2y)(2)\hat{j}$$

$$\vec{a}_{(1,2,3)} = 1\hat{i} + 2(2)\hat{j}$$

$$= 1\hat{i} + 8\hat{j}$$

$$|\vec{a}| = \sqrt{1^2 + 8^2}$$

$$= \sqrt{65}$$

Question: Angular momentum of the electron in a hydrogen atom is $\frac{3h}{2\pi}$, then find the energy of the electron in _____ ev.

Options:

- (a) - 1.51 ev
- (b) - 2.66 ev
- (c) - 0.52 ev
- (d) - 1.00 ev

Answer: (a)

Solution: (a)

$$L = \frac{nh}{2\pi} = \frac{3h}{2\pi} \Rightarrow n = 3$$

$$E = -13.6 \frac{z^2}{n^2}$$

$$= -13.6 \frac{(1)^2}{3^2} = -\frac{13.6}{9} = -1.51 \text{ ev}$$

Question: The equation of a plane progressive wave is given by

$y = 5 \cos \pi \left(200t - \frac{x}{150} \right)$. Where x & y are in cm and t is in seconds. Find the wave velocity for the given wave function.

Options:

- (a) 100 m/s
- (b) 200 m/s
- (c) 300 m/s
- (d) 400 m/s

Answer: (c)

Solution:

$$V = \frac{\omega}{k} = \frac{200\pi}{\left(\frac{\pi}{150}\right)} \text{ cm/s}$$

$$= 200 \times 150 \times 10^{-2} \text{ m/s}$$

$$= 300 \text{ m/s}$$

Question: For Diatomic gas, find the ration of $\Delta Q : \Delta U : W$ for isobaric process

Options:

- (a) 2:5:7
- (b) 2:3:5
- (c) 2:7:3
- (d) 7:5:2

Answer: (d)

Solution:

$$\Delta Q = nC_p \Delta T$$

$$\Delta U = nC_v \Delta T$$

$$W = nR \Delta T$$

$$\Delta Q : \Delta U : W = C_p : C_v : R$$

$$= \frac{7R}{2} : \frac{5R}{2} : R$$

$$= 7 : 5 : 2$$

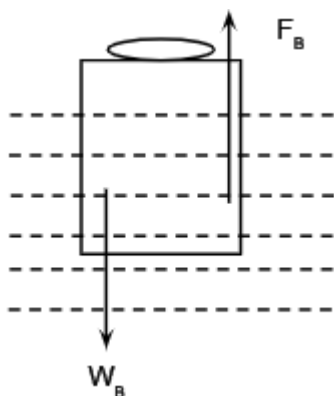
Question: A wooden cubic block of relative density. 0.4 is floating in water. Side of cubic block is 10 cm. When a coin is placed on the block the dips 0.3 cm in equilibrium. Weight of coin is

Options:

- (a) 0.3N
- (b) 230 N
- (c) 0.7 N
- (d) 3N

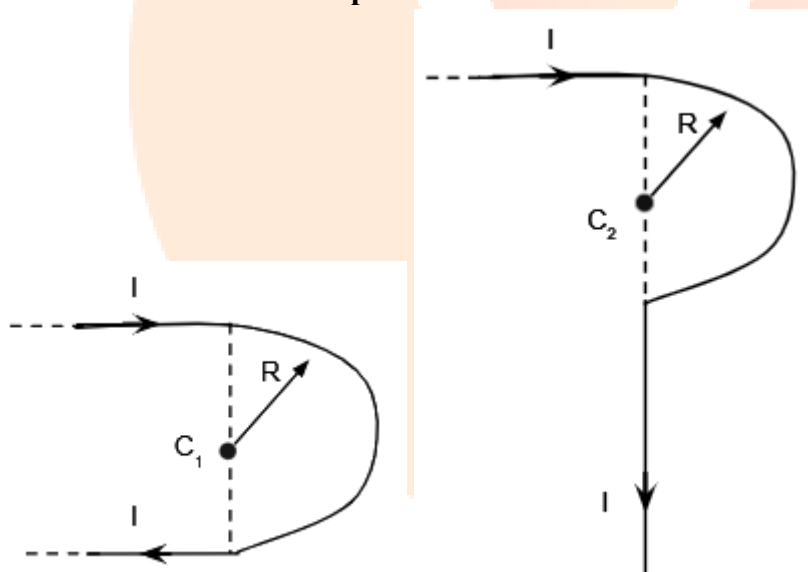
Answer: (a)

Solution:



$$\begin{aligned}
 W_{coin} &= \Delta F_B \\
 &= \rho_W (V_{disp}) g \\
 &= 10^3 (10 \times 10 \times 10^{-4} \times 0.3 \times 10^{-2}) \times 10 \\
 &= 0.3 \text{ N}
 \end{aligned}$$

Question: Consider two arrangements of wires, find ratio of magnetic field at centre of the semi-circular part.



Options:

- $\frac{\pi + 3}{\pi + 4}$
 (a) $\frac{\pi - 1}{\pi + 4}$
 (b) $\frac{\pi + 2}{\pi + 2}$
 (c) $\frac{\pi + 1}{\pi + 1}$

$$\frac{\pi - 1}{\pi + 1}$$

(d) $\pi + 1$

Answer: (c)

Solution:

$$\frac{\mu_0 i}{4\pi R} \times 2 + \frac{1}{2} \left(\frac{\mu_0 i}{2R} \right) = B_1$$

$$B_2 = \frac{\mu_0 i}{4\pi R} + \frac{1}{2} \left(\frac{\mu_0 i}{2R} \right)$$

Question: Weight of string = w and weight of block = w and cross sectional area of

string is A if stress at distance L/3 from top and is given by $\frac{w}{A} + \frac{2W}{\gamma A}$ determine y ?



Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (c)

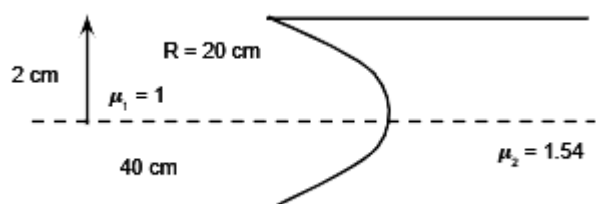
Solution:

$$= \frac{\frac{2w}{3} + w}{A} = \frac{2W}{3A} + \frac{w}{A}$$

$$= \frac{\frac{5w}{3}}{A} = \text{Stress}$$

$$y = 3$$

Question: Determine height of image after refraction.



Options:

- (a) 25 cm
- (b) -25 cm
- (c) 40 cm
- (d) -40 cm

Answer: (b)

Solution:



Step 1: Find Image Position (v)

Using the refraction formula for a spherical surface:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Substituting the values:

✓

$$\frac{1.54}{v} - \frac{1}{-40} = \frac{1.54 - 1}{20}$$

$$\frac{1.54}{v} + 0.025 = \frac{0.54}{20}$$

$$\frac{1.54}{v} = 0.027 - 0.025 = 0.002$$

$$v = \frac{1.54}{0.002} = 770 \text{ cm}$$

Step 2: Find Height of Image (h_I)

The lateral magnification (m) for refraction at a spherical surface is:

$$m = \frac{h_I}{h_O} = \left(\frac{\mu_1}{\mu_2} \right) \left(\frac{v}{u} \right)$$

Substituting the values:

✓

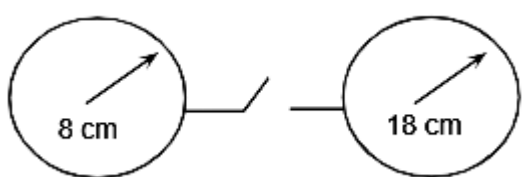
$$\frac{h_I}{2} = \left(\frac{1}{1.54} \right) \left(\frac{770}{-40} \right)$$

$$\frac{h_I}{2} = \left(\frac{1}{1.54} \right) \times (-19.25)$$

$$\frac{h_I}{2} = -12.5$$

$$h_I = -25 \text{ cm}$$

Question: Ratio of Electric field intensity on surface of sphere radius 18cm to that on the surface of sphere of radius 8 cm after closing of switch till equilibrium Established



Options:

- (a) 2:7
- (b) 7:2
- (c) 4:9
- (d) 9:4

Answer: (c)

Solution:

Solution**1. Equilibrium Condition:**

When two conducting spheres are connected by a wire and a switch is closed, charge flows until they reach the same **electric potential** (V).

✓

$$V_1 = V_2$$

$$\frac{kQ_1}{R_1} = \frac{kQ_2}{R_2} \implies \frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

2. Electric Field Intensity (E):

The electric field on the surface of a sphere is given by:

$$E = \frac{kQ}{R^2}$$

3. Ratio of Electric Fields:

Let $R_1 = 18$ cm and $R_2 = 8$ cm. The ratio of their electric fields is:

$$\frac{E_1}{E_2} = \frac{\frac{kQ_1}{R_1^2}}{\frac{kQ_2}{R_2^2}} = \left(\frac{Q_1}{Q_2} \right) \left(\frac{R_2^2}{R_1^2} \right)$$

From the potential equilibrium condition ($\frac{Q_1}{Q_2} = \frac{R_1}{R_2}$), we substitute:

$$\frac{E_1}{E_2} = \left(\frac{R_1}{R_2} \right) \left(\frac{R_2^2}{R_1^2} \right) = \frac{R_2}{R_1}$$

4. Final Calculation:

$$\frac{E_1}{E_2} = \frac{8}{18} = \frac{4}{9}$$

Question: Capacitor with plate separation of d filled with air has some capacitance now dielectric slab of $K = 5$ of thickness $d/2$ is inserted determine % creasing in capacitance.



Options:

- (a) 70%
- (b) 66.66%
- (c) 33.33%
- (d) 10%

Answer: (b)

Solution:

Solution**1. Initial Capacitance (C_0):**

For an air-filled capacitor with area A and separation d :

$$C_0 = \frac{\epsilon_0 A}{d}$$

2. New Capacitance with Dielectric (C_{new}):

When a slab of thickness t and dielectric constant K is inserted, the new capacitance is given by:

$$C_{new} = \frac{\epsilon_0 A}{d - t + \frac{t}{K}}$$

3. Substitute the given values ($t = d/2$ and $K = 5$):

$$C_{new} = \frac{\epsilon_0 A}{d - \frac{d}{2} + \frac{d/2}{5}}$$

$$C_{new} = \frac{\epsilon_0 A}{\frac{d}{2} + \frac{d}{10}}$$

$$C_{new} = \frac{\epsilon_0 A}{\frac{5d+d}{10}} = \frac{10\epsilon_0 A}{6d} = \frac{5}{3}C_0$$

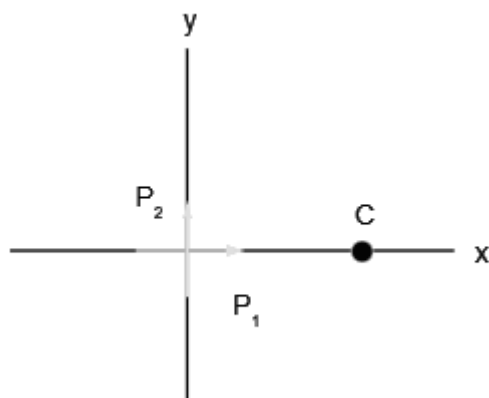
4. Calculate Percentage Increase:

$$\text{Percentage Increase} = \left(\frac{C_{new} - C_0}{C_0} \right) \times 100$$

$$\text{Percentage Increase} = \left(\frac{\frac{5}{3}C_0 - C_0}{C_0} \right) \times 100 = \frac{2}{3}$$

Answer: The capacitance increases by approximately 66.67%.

Question: If at point c net electric field intensity is at angle 60° with x axis. Determine P_2/P_1 .



Options:

- (a) $2\sqrt{5}$
- (b) $3\sqrt{5}$
- (c) $2\sqrt{7}$
- (d) $2\sqrt{3}$

Answer: (d)

Solution:

The diagram shows two dipoles, P_1 (along the x-axis) and P_2 (along the y-axis). Point C is located on the x-axis.

1. **Electric Field due to P_1 at point C :**

Since point C is on the axial line of dipole P_1 :

$$E_1 = \frac{2kP_1}{r^3}$$

(directed along the +x axis) ✓

2. **Electric Field due to P_2 at point C :**

Since point C is on the equatorial line of dipole P_2 :

$$E_2 = \frac{kP_2}{r^3}$$

(directed along the +y axis) ✓

3. **Net Field Angle:**

The problem states the net field E_{net} makes an angle $\theta = 60^\circ$ with the x-axis. Therefore:

$$\tan(60^\circ) = \frac{E_y}{E_x} = \frac{E_2}{E_1}$$

4. Calculate the Ratio:

Substitute the expressions for E_1 and E_2 :

$$\sqrt{3} = \frac{\left(\frac{kP_2}{r^3}\right)}{\left(\frac{2kP_1}{r^3}\right)}$$

v

$$\sqrt{3} = \frac{P_2}{2P_1}$$

v

$$\frac{P_2}{P_1} = 2\sqrt{3}$$

Question: For an ideal gas having $C_p = 3R$ and $C_v = 2R$. Find work done by one mole of the gas in adiabatic expansion when pressure reduces from 8 bar to 1 bar. (Initial temperature = 140°C)

Options:

- (a) 1140 R
- (b) 270 R
- (c) 826 R
- (d) 413 R

Answer: (d)

Solution:

1. Identify Constants:

- $\gamma = \frac{C_p}{C_v} = \frac{3R}{2R} = 1.5$
- Initial Temperature (T_1) in Kelvin: $140 + 273 = 413 \text{ K}$
- $P_1 = 8 \text{ bar}$, $P_2 = 1 \text{ bar}$, $n = 1 \text{ mole}$

2. Find Final Temperature (T_2):

Using the adiabatic relation between P and T :

$$T \cdot P^{\frac{1-\gamma}{\gamma}} = \text{constant} \implies T_1 P_1^{\frac{1-\gamma}{\gamma}} = T_2 P_2^{\frac{1-\gamma}{\gamma}}$$

$$T_2 = T_1 \left(\frac{P_1}{P_2} \right)^{\frac{1-\gamma}{\gamma}}$$

Substitute the values:

$$T_2 = 413 \times \left(\frac{8}{1} \right)^{\frac{1-1.5}{1.5}} = 413 \times (8)^{-\frac{0.5}{1.5}} = 413 \times (8)^{-1/3}$$

$$T_2 = \frac{413}{\sqrt[3]{8}} = \frac{413}{2} = 206.5 \text{ K}$$

3. Calculate Work Done (W):

For an adiabatic process:

$$W = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

$$W = \frac{1 \cdot R \cdot (413 - 206.5)}{1.5 - 1}$$

$$W = \frac{R \cdot 206.5}{0.5} = 206.5 \times 2 \times R$$

$$W = 413R$$

Question: Photons of wavelength λ & 3λ incident on a metal surface if stopping potential for the ejected photoelectrons are $4V_0$ & V_0 respectively. Find threshold wavelength.

Options:

- (a) 6λ
- (b) 8λ
- (c) 9λ
- (d) 2λ

Answer: (c)

Solution:

Solution

Using the photoelectric equation:

$$K_{max} = eV_s = \frac{hc}{\lambda} - \phi$$

Where $\phi = \frac{hc}{\lambda_0}$ is the work function and λ_0 is the threshold wavelength.

For the first case:

$$4eV_0 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \quad \dots (\text{Equation 1})$$

For the second case:

$$eV_0 = \frac{hc}{3\lambda} - \frac{hc}{\lambda_0} \quad \dots (\text{Equation 2})$$

Step 1: Eliminate eV_0

Multiply Equation 2 by 4:

$$4eV_0 = \frac{4hc}{3\lambda} - \frac{4hc}{\lambda_0}$$

Step 2: Equate the two expressions for $4eV_0$

$$\frac{hc}{\lambda} - \frac{hc}{\lambda_0} = \frac{4hc}{3\lambda} - \frac{4hc}{\lambda_0}$$

Step 3: Solve for λ_0

Divide through by hc :

$$\frac{1}{\lambda} - \frac{1}{\lambda_0} = \frac{4}{3\lambda} - \frac{4}{\lambda_0}$$

Rearrange terms containing λ_0 on one side:

$$\frac{4}{\lambda_0} - \frac{1}{\lambda_0} = \frac{4}{3\lambda} - \frac{1}{\lambda}$$

$$\frac{3}{\lambda_0} = \frac{4 - 3}{3\lambda}$$

$$\frac{3}{\lambda_0} = \frac{1}{3\lambda}$$

$$\lambda_0 = 9\lambda$$

Question: in YDSE experiment wavelength of light used is 620 nm and separation between slits is 0.2 mm. Find angular fringe width.

Options:

- (a) 3×10^{-4}
- (b) 6.2×10^{-4}
- (c) 3.1×10^{-3}
- (d) 1.2×10^{-3}

Answer: (c)

Solution:

$$\theta = \frac{\lambda}{d}$$

Where:

- λ is the wavelength of light.
- d is the distance between the slits.

Step 1: Convert units to meters

- $\lambda = 620 \text{ nm} = 620 \times 10^{-9} \text{ m}$
- $d = 0.2 \text{ mm} = 0.2 \times 10^{-3} \text{ m} = 2 \times 10^{-4} \text{ m}$

Step 2: Calculate the value

$$\theta = \frac{620 \times 10^{-9}}{2 \times 10^{-4}}$$

$$\theta = 310 \times 10^{-5}$$

$$\theta = 3.1 \times 10^{-3} \text{ radians}$$

Question: A particle is rotating in a circular path and at any instance it's position θ and

time t are related as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$, then the angular acceleration after 10 seconds is

Answer: (130)

Solution:

1. Given Equation

The angular position θ is given by:

$$\theta = \frac{5t^4}{40} - \frac{t^3}{3}$$

This simplifies to:

$$\theta = \frac{1}{8}t^4 - \frac{1}{3}t^3$$

2. Find Angular Velocity (ω)

The angular velocity is the first derivative of θ :

$$\omega = \frac{d\theta}{dt} = \frac{d}{dt} \left(\frac{1}{8}t^4 - \frac{1}{3}t^3 \right)$$

$$\omega = \frac{4}{8}t^3 - \frac{3}{3}t^2 = 0.5t^3 - t^2$$

3. Find Angular Acceleration (α)

The angular acceleration is the derivative of ω :

$$\alpha = \frac{d\omega}{dt} = \frac{d}{dt}(0.5t^3 - t^2)$$

$$\alpha = (0.5 \times 3)t^2 - 2t$$

$$\alpha = 1.5t^2 - 2t$$

4. Calculate at $t = 10$ seconds

Substitute $t = 10$ into the acceleration formula:

$$\alpha = 1.5(10)^2 - 2(10)$$

$$\alpha = 1.5(100) - 20$$

$$\alpha = 150 - 20 = 130 \text{ rad/s}^2$$

Question: A liquid drop of diameter 2mm, breaks into 512 droplets. The change in the surface energy is $a \times 10^{-6}$, the value of a is ____ (take the surface tension of the liquid = 0.08N/m)

Answer: (7)

Solution:

1. Relationship Between Radii

- **Initial Radius (R):** 1 mm (since diameter is 2 mm) = 10^{-3} m
- **Number of droplets (n):** 512
- By conservation of volume: $R^3 = n \cdot r^3$

$$R = \sqrt[3]{512} \cdot r = 8r \implies r = \frac{R}{8}$$

2. Change in Surface Area (ΔA)

- Initial Area (A_1): $4\pi R^2$
- Final Area (A_2): $512 \cdot (4\pi r^2) = 512 \cdot 4\pi \left(\frac{R}{8}\right)^2 = 512 \cdot \frac{4\pi R^2}{64} = 8 \cdot (4\pi R^2)$
- ΔA : $8(4\pi R^2) - 4\pi R^2 = 7(4\pi R^2) = 28\pi R^2$

3. Change in Surface Energy (ΔU)

- $\Delta U = T \cdot \Delta A$
- Substitute $T = 0.08$ N/m and $R = 10^{-3}$ m:

$$\Delta U = 0.08 \cdot 28 \cdot \pi \cdot (10^{-3})^2$$

$$\Delta U = 2.24 \cdot \pi \cdot 10^{-6}$$

$$\Delta U \approx 7.03 \times 10^{-6} \text{ J}$$

JEE-Main-02-04-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: The correct increasing order of bond length among the following is

Options:

- (a) O_2^+ , O_2 , O_2^- , O_2^{2-}
- (b) O_2^+ , O_2 , O_2^{2-} , O_2^-
- (c) O_2^{2-} , O_2^- , O_2 , O_2^+
- (d) O_2^- , O_2^{2-} , O_2^+ , O_2

Answer: (a)

Solution: (a) O_2^+ , O_2 , O_2^- , O_2^{2-}

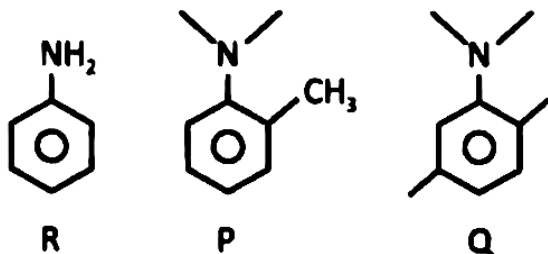
Question: For first order reaction, rate constant at 27°C and $t^\circ\text{C}$ is 1.5×10^3 and 4.5×10^3 respectively. If the activation energy of reaction is 60 kJ then find temperature t.

Options:

Answer: (41)

Solution:

Question: Write the correct order of rate of reaction of following with PhN_2Cl



Options:

- (a) $R > P > Q$
- (b) $P > R > Q$
- (c) $Q > P > R$
- (d) $P > Q > R$

Answer: (a)

Solution: (a) $R > P > Q$

Question: Match List-I with List-II

List-I	List-II
I) Vitamin C	A) Thiamine
II) Vitamin B ₁	B) Riboflavin

III) Vitamin B ₆	C) Ascorbic Acid
IV) Vitamin B ₂	D) Pyridoxine

Options:

- (a) I-A, II-B, III-C, IV-D
- (b) I-C, II-A, III-D, IV-B
- (c) I-A, II-C, III-B, IV-D
- (d) I-C, II-D, III-A, IV-B

Answer: (b)

Solution: (b) I-C, II-A, III-D, IV-B

Question: Calculate number of isomers of C₅H₁₀, which does not decolourise cold alkaline KMnO₄ solution

Options:

Answer: (7)

Solution:

Question: Arrange following functional group in the increasing order of priority of functional group:

-CONH₂, -C=O, -CHO, -CN, -C≡C, -NH₂

(1) (2) (3) (4) (5) (6)

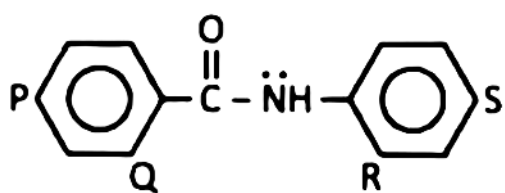
Options:

- (a) 1 > 4 > 2 > 3 > 6 > 5
- (b) 1 > 4 > 2 > 3 > 5 > 6
- (c) 1 > 4 > 3 > 2 > 6 > 5
- (d) 1 > 2 > 4 > 3 > 6 > 5

Answer: (c)

Solution:

Question: In the given compound, the electrophile attack will be favoured at



Options:

- (a) S
- (b) P
- (c) Q
- (d) R

Answer: (a)

Solution:

Question: Which compound gives apple green colour in flame test?

Options:

- (a) BaSO₄
- (b) Mn₃(PO₄)₂

(c) $\text{Ca}_3(\text{PO}_4)_2$

(d) $\text{Al}_2(\text{SO}_4)_3$

Answer: (a)

Solution: (a) BaSO_4

Question: One mole of alkane reacts with 8 mole of O_2 . The molecular formula of alkane is

Options:

(a) C_4H_{10}

(b) C_5H_{12}

(c) C_2H_{14}

(d) C_3H_8

Answer: (b)

Solution: (b) C_5H_{12}

Question: 18g of H_2O reacts with x g of Fe to produce Fe_2O_3 , 'x'. Is, (Atomic mass of Fe = 56 u)

Options:

(a) 84 g

(b) 63 g

(c) 42 g

(d) 51 g

Answer: (c)

Solution: (c) 42 g

Question: The K_{sp} values of Ag_2CrO_4 and AgBr are $32x$ and $4y$ respectively. The ratio of solubilities $\frac{S_1}{S_2}$ is

[Here, S_1 is solubility of Ag_2CrO_4 and S_2 is solubility of AgBr]

Options:

(a) $\frac{2x^{\frac{1}{3}}}{y^{\frac{1}{2}}}$

(b) $\frac{x^{\frac{1}{3}}}{y^{\frac{1}{2}}}$

(c) $\frac{x^{\frac{1}{3}}}{2y^{\frac{1}{2}}}$

(d) $\frac{(2x)^{\frac{1}{3}}}{(2y)^{\frac{1}{2}}}$

Answer: (b)

Solution: (b) $\frac{x^{\frac{1}{3}}}{y^{\frac{1}{2}}}$

Question: The orbit angular momentum of e in nth shell of H atom is $\frac{3h}{\pi}$. The total energy of e is

Options:

(a) -0.38 eV

- (b) -3.4 eV
(c) -0.544 eV
(d) -0.85 eV

Answer: (a)

Solution: (a) -0.38 eV

Question: Arrange the following complexes in the increasing order of crystal field splitting energy (Δ_0)

- (a) $[\text{Cr}(\text{CN})_6]^{3-}$ (b) $[\text{CrF}_6]^{3-}$
(c) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Cr}(\text{en})_3]^{3+}$

Options:

- (a) $b < c < d < a$
(b) $a < b < c < d$
(c) $a < d < c < b$
(d) $b < a < c < d$

Answer: (a)

Solution: (a) $b < c < d < a$

Question: Calculate the energy required for the following process $\text{Li}^{2+} \rightarrow \text{Li}^{3+} + e^-$

Given: Ground state energy of Hydrogen is -13.6 eV/atom

Options:

- (a) 13.6 eV/atom
(b) 122.4 eV/atom
(c) 54.4 eV/atom
(d) 30.6 eV/atom

Answer: (b)

Solution: (b) 122.4 eV/atom

Question: 20 g fluoroacetic acid is dissolved in 500 gm water. If depression in freezing point is 1°C , then calculate K_a for fluoroacetic acid. [Assume molality is same as molarity]

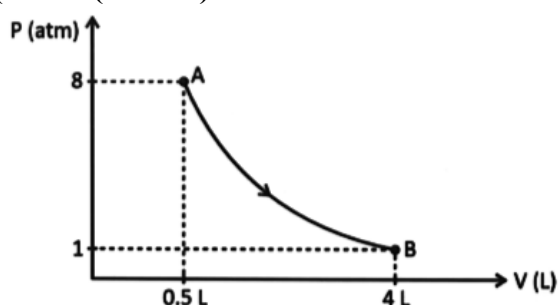
Options:

- (a) 1.18×10^{-3}
(b) 1.5×10^{-5}
(c) 1.18×10^{-4}
(d) 1.2×10^{-6}

Answer: (a)

Solution: (a) 1.18×10^{-3}

Question: Calculate work in following process during A to B, in following graph. (Given ($V = 3R$))



Options:

- (a) -6.2 L.atm
- (b) -8.3 L.atm
- (c) -4.8 L.atm
- (d) -9.2 L.atm

Answer: (b)

Solution: (b) -8.3 L.atm

Question: Match List-I with List-II

List-I (Molecules)	List-II (Test)
I) Ethanol	P) Neutral FeCl_3
II) Phenol	Q) Ceric ammonium nitrate
III) Ethanoic Acid	R) Schiff reagent
IV) Acetaldehyde	S) NaHCO_3

Options:

- (a) I-P, II-Q, III-R, IV-S
- (b) I-Q, II-S, III-P, IV-R
- (c) I-Q, II-P, III-S, IV-S
- (d) I-P, II-Q, III-S, IV-R

Answer: (c)

Solution: (c) I-Q, II-P, III-S, IV-S

Question: Which of the following relation about first order and zero order reaction is correct

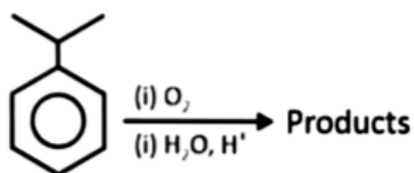
Options:

- (a) $(t_{1/2})_{\text{first order}} = 2(t_{100\%})_{\text{first order}}$
- (b) $(t_{100\%})_{\text{first order}} = \text{Infinite time of } t_{1/2} \text{ of first order}$
- (c) $(t_{1/2})_{\text{zero order}} = 2 \times (t_{100\%})_{\text{zero order}}$
- (d) $(t_{100\%})_{\text{zero order}} = \text{Infinite time of } t_{1/2} \text{ of zero order}$

Answer: (b)

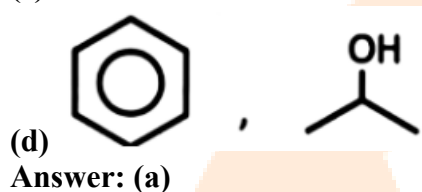
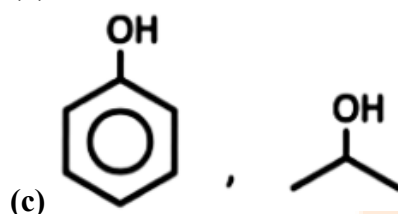
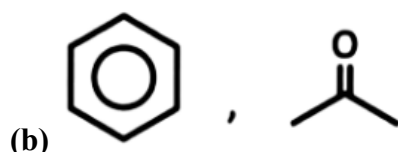
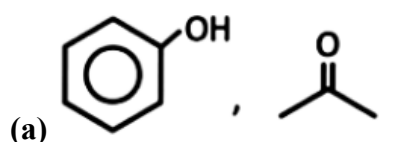
Solution: (b) $(t_{100\%})_{\text{first order}} = \text{Infinite time of } t_{1/2} \text{ of first order}$

Question: Consider the following reaction:

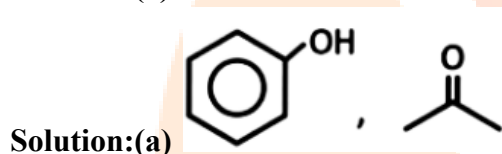


Products obtained are

Options:



Answer: (a)



Question: Given below are two statements.

Statement-I: First I.E. order : $\text{Na} > \text{Mg} > \text{Al}$

Statement-II: 3rd I.E. order : $\text{Ca} > \text{Al} > \text{Na}$.

Options:

- (a) Both S-I and S-II are incorrect
- (b) S-I is correct but S-II is incorrect
- (c) S-I is incorrect but S-II is correct
- (d) Both S-I and S-II are correct

Answer: (a)

Solution: (a) Both S-I and S-II are incorrect

Question: 8 g of $[\text{CrCl}_x(\text{H}_2\text{O})_y]\text{Cl}_{(3-x)} \times \text{H}_2\text{O}_{(6-y)}$ reacts with average AgNO_3 to form 8.61 g of AgCl . The value of $(x \times y)$ is ____.

Options:

Answer: (5)

JEE-Main-02-04-2026 (Memory Based)

[MORNING SHIFT]

Maths

Question: Let $\sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18} = k$. Then find the value of $\sin 10k \frac{\pi}{3}$

Solution:

Step 1: Find the value of k

The given expression for k is:

$$k = \sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18}$$

Converting the angles to degrees to make them easier to work with ($\pi = 180^\circ$):

- $\frac{\pi}{18} = 10^\circ$
- $\frac{5\pi}{18} = 50^\circ$
- $\frac{7\pi}{18} = 70^\circ$

So, $k = \sin 10^\circ \sin 50^\circ \sin 70^\circ$.

We can use the standard trigonometric identity:

We can use the standard trigonometric identity:

$$\sin \theta \sin(60^\circ - \theta) \sin(60^\circ + \theta) = \frac{1}{4} \sin 3\theta$$

By setting $\theta = 10^\circ$, we get:

$$k = \sin 10^\circ \sin(60^\circ - 10^\circ) \sin(60^\circ + 10^\circ) = \frac{1}{4} \sin(3 \times 10^\circ)$$

$$k = \frac{1}{4} \sin 30^\circ = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

Step 2: Calculate the required value

Now we need to find the value of $\sin \left(10k \frac{\pi}{3}\right)$.

Substitute $k = \frac{1}{8}$ into the expression:

Value = $\sin\left(10 \times \frac{1}{8} \times \frac{\pi}{3}\right)$

Value = $\sin\left(\frac{5}{4} \times \frac{\pi}{3}\right) = \sin\frac{5\pi}{12}$

Converting $\frac{5\pi}{12}$ to degrees:

$$\frac{5 \times 180^\circ}{12} = 5 \times 15^\circ = 75^\circ$$

We know that $\sin 75^\circ = \sin(45^\circ + 30^\circ)$, which simplifies to:

$$\sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

Final Answer:

The value is $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (which can also be written as $\frac{\sqrt{6}+\sqrt{2}}{4}$).

Question: $(1 + \sin x) \frac{dy}{dx} + \cos x = 0$

Solution:

Step 1: Separate the Variables

Rearrange the equation to move all terms involving y to one side and all terms involving x to the other:

$$(1 + \sin x) \frac{dy}{dx} = -\cos x$$

$$dy = -\frac{\cos x}{1 + \sin x} dx$$

Step 2: Integrate Both Sides

Now, we integrate:

$$\int dy = - \int \frac{\cos x}{1 + \sin x} dx$$

To solve the right-hand side, use **substitution**:

- Let $u = 1 + \sin x$
- Then $du = \cos x dx$

The integral becomes:

$$y = - \int \frac{1}{u} du$$

$$y = -\ln|u| + C$$

Step 3: Final Solution

Substitute back $u = 1 + \sin x$:

$$y = -\ln|1 + \sin x| + C$$

Question: Find the range of $\sin^{-1}\left(\frac{1}{x^2 - 2x + 2}\right)$

Solution:

• **Step 1: Analyze the Argument**

The quadratic $x^2 - 2x + 2$ can be written as $(x - 1)^2 + 1$.

- Minimum value = 1 (at $x = 1$).
- Maximum value = ∞ .

Therefore, the range of the denominator is $[1, \infty)$.

• **Step 2: Range of the Fraction**

The range of $g(x) = \frac{1}{[1, \infty)}$ is $(0, 1]$.

• **Step 3: Apply $\sin^{-1}$**

The function $\sin^{-1}(t)$ is increasing.

- $\sin^{-1}(\text{approaching } 0) \rightarrow 0$
- $\sin^{-1}(1) = \pi/2$

• **Final Range: $(0, \frac{\pi}{2}]$**

Question: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (3, 4) is a point, $e = \frac{\sqrt{5}}{3}$ Find length of Latus Rectum
Solution:

• **Step 1: Use Eccentricity to relate a and b**

For an ellipse, $e^2 = 1 - \frac{b^2}{a^2}$.

Given $e = \frac{\sqrt{5}}{3} \implies e^2 = \frac{5}{9}$.

$$\frac{5}{9} = 1 - \frac{b^2}{a^2} \implies \frac{b^2}{a^2} = 1 - \frac{5}{9} = \frac{4}{9}$$

Therefore, $b^2 = \frac{4}{9}a^2$.

• **Step 2: Substitute the point (3, 4) into the equation**

Since (3, 4) lies on the ellipse:

Substitute $b^2 = \frac{4}{9}a^2$:

$$\frac{9}{a^2} + \frac{16}{\frac{4}{9}a^2} = 1 \implies \frac{9}{a^2} + \frac{36}{a^2} = 1$$

$$\frac{45}{a^2} = 1 \implies a^2 = 45$$

• **Step 3: Calculate b^2 and a**

$$b^2 = \frac{4}{9} \times 45 = 20$$

$$a = \sqrt{45} = 3\sqrt{5}$$

• **Step 4: Find the Length of Latus Rectum (LR)**

The formula for LR is $\frac{2b^2}{a}$:

$$LR = \frac{2 \times 20}{3\sqrt{5}} = \frac{40}{3\sqrt{5}}$$

Rationalizing the denominator:

$$LR = \frac{40\sqrt{5}}{3 \times 5} = \frac{8\sqrt{5}}{3}$$

Question: The number of seven digit numbers, that can be formed by using the digits 1, 2, 3, 5, 7 such that each digit is used at least once is

Options:

- (a) 16800
- (b) 15400
- (c) 29400
- (d) 17800

Answer: (a)

Solution:

$${}^5C_1 \times \frac{7!}{3!} = 5 \times 7 \times 6 \times 5 \times 4 = 4200$$

$${}^5C_2 \times \frac{7!}{2!2!} = 10 \times 1260 = \frac{12600}{16800}$$

$$x \in \sum -\pi, \pi$$

Question: Find number of solutions of equations in $-\pi$ to π $\sin x(\sin x + \cos x) = a$ where a is an integer

Solution:

- **Step 1: Simplify the expression**

$$\text{Let } f(x) = \sin x(\sin x + \cos x)$$

$$f(x) = \sin^2 x + \sin x \cos x$$

Using the double angle identities $\sin^2 x = \frac{1 - \cos 2x}{2}$ and $\sin x \cos x = \frac{\sin 2x}{2}$:

$$f(x) = \frac{1 - \cos 2x}{2} + \frac{\sin 2x}{2} = \frac{1}{2} + \frac{1}{2}(\sin 2x - \cos 2x)$$

- **Step 2: Find the Range of the function**

The expression $(\sin 2x - \cos 2x)$ can be written as $\sqrt{2} \sin(2x - \pi/4)$.

The range of $\sqrt{2} \sin(2x - \pi/4)$ is $[-\sqrt{2}, \sqrt{2}]$.

Substituting this back:

$$\text{Range of } f(x) = \left[\frac{1 - \sqrt{2}}{2}, \frac{1 + \sqrt{2}}{2} \right]$$

Approximating the values ($\sqrt{2} \approx 1.414$):

$$\text{Range} \approx \left[\frac{1 - 1.414}{2}, \frac{1 + 1.414}{2} \right] \approx [-0.207, 1.207]$$

- **Step 3: Identify Integer values of a**

Since a must be an integer within this range, the possible values for a are:

$$a = 0 \text{ and } a = 1.$$

- **Step 4: Count the Solutions in $(-\pi, \pi)$**

The period of $f(x)$ is π because it depends on $2x$. In the interval $(-\pi, \pi)$, which has a length of 2π , we will see **two full periods** of the graph.

1. **For $a = 0$:** The value 0 is within the range but is not an extreme point. In one period (π) , $f(x) = 0$ occurs twice. In $(-\pi, \pi)$, there are **4 solutions**.
2. **For $a = 1$:** The value 1 is also within the range but not an extreme point. In one period, $f(x) = 1$ occurs twice. In $(-\pi, \pi)$, there are **4 solutions**.

Question: Two APS with a,d,n as 1,5,101 and 9,7,71. No. of common terms

Solution:

Step 1: Write the general terms of both APs

- AP 1: $a_1 = 1, d_1 = 5, n_1 = 101$
Last term (l_1) = $1 + (101 - 1)5 = 1 + 500 = 501$
- AP 2: $a_2 = 9, d_2 = 7, n_2 = 71$
Last term (l_2) = $9 + (71 - 1)7 = 9 + 490 = 499$

The common terms must be less than or equal to the smaller of the two last terms: **Common Terms** ≤ 499 .

Step 2: Find the first common term (a_{common})

List the terms of both sequences until a match is found:

- AP 1: 1, 6, 11, 16, 21, 26, 31, 36, ...
- AP 2: 9, 16, 23, 30, 37, ...

The first common term is 16.

Step 3: Find the Common Difference (D)

The common terms of two APs also form an AP. Its common difference is the **LCM** of the individual common differences:

$$D = \text{LCM}(d_1, d_2) = \text{LCM}(5, 7) = 35$$

Step 4: Find the number of terms (n)

The sequence of common terms is: 16, 51, 86, ...

We use the formula $a + (n - 1)D \leq \text{Max Limit}$:

$$16 + (n - 1)35 \leq 499$$

$$(n - 1)35 \leq 483$$

$$n - 1 \leq \frac{483}{35}$$

$$n - 1 \leq 13.8$$

$$n \leq 14.8$$

Since n must be an integer, the number of common terms is 14.

Question: If $L_1 : \frac{x - a}{2} = \frac{y - 2}{3} = \frac{z - b}{6}$ and $L_2 : \frac{x - b}{3} = \frac{y - 7}{6} = \frac{z - 1}{3}$ intersect in xy plane. Then the value of $|a + b|$ is

Solution:

- **Step 1: Use the xy-plane condition**

Since the lines intersect in the xy-plane, the z-coordinate of the intersection point must be 0.

Substitute $z = 0$ into both line equations to find the intersection point parameters.

- **Step 2: Solve for the parameters**

For L_1 at $z = 0$:

$$\frac{x-a}{2} = \frac{y-2}{3} = \frac{0-b}{6} = -\frac{b}{6}$$

From this, we get:

$$x = a - \frac{b}{3} \quad \text{and} \quad y = 2 - \frac{b}{2}$$

For L_2 at $z = 0$:

$$\frac{x-b}{3} = \frac{y-7}{6} = \frac{0-1}{3} = -\frac{1}{3}$$

From this, we get:

$$x - b = 3 \left(-\frac{1}{3} \right) \Rightarrow x = b - 1$$

$$y - 7 = 6 \left(-\frac{1}{3} \right) \Rightarrow y = 7 - 2 = 5$$

- **Step 3: Equate x and y to find a and b**

Comparing y -coordinates:

$$2 - \frac{b}{2} = 5 \Rightarrow -\frac{b}{2} = 3 \Rightarrow b = -6$$

Comparing x -coordinates:

$$a - \frac{b}{3} = b - 1$$

Substitute $b = -6$:

$$a - \left(\frac{-6}{3} \right) = -6 - 1$$

$$a + 2 = -7 \Rightarrow a = -9$$

- **Step 4: Final Calculation**

$$|a + b| = |(-9) + (-6)| = |-15| = 15$$

Question: ${}^{36}C_{r+1} = 6 \cdot \frac{{}^{35}C_r}{k^2 - 3}$ Such that $S = \{(r, k)\}$ then number of elements in set S is

Solution:

• **Step 1: Simplify the Combinations**

Use the property: ${}^{36}C_{r+1} = \frac{36}{r+1} \cdot {}^{35}C_r$.

Substitute this into the original equation:

$$\frac{36}{r+1} \cdot {}^{35}C_r = \frac{6}{k^2-3} \cdot {}^{35}C_r$$

Canceling ${}^{35}C_r$ from both sides (since it's non-zero):

$$\frac{36}{r+1} = \frac{6}{k^2-3}$$

$$\frac{6}{r+1} = \frac{1}{k^2-3} \implies k^2-3 = \frac{r+1}{6}$$

$$k^2 = 3 + \frac{r+1}{6}$$

• **Step 2: Identify Constraints for r**

For ${}^{36}C_{r+1}$ to be defined, $0 \leq r+1 \leq 36$.

Therefore, $-1 \leq r \leq 35$.

Since $(r+1)$ must be a multiple of 6 for k^2 to potentially be an integer (and for the fraction to simplify nicely), let's check values for $r+1$:

$r+1 \in \{6, 12, 18, 24, 30, 36\}$.

• **Step 3: Test for Integer k**

$|r+1| \mid |k^2-3| = \frac{r+1}{6} \mid k$ (Integer?) | Pairs (r, k) |

| :--- | :--- | :--- | :--- | :--- |

| 6 | 5 | $3+1=4 \mid \pm 2 \mid (5, 2), (5, -2)$ |

| 12 | 11 | $3+2=5 \mid \pm \sqrt{5}$ (No) | - |

| 18 | 17 | $3+3=6 \mid \pm \sqrt{6}$ (No) | - |

| 24 | 23 | $3+4=7 \mid \pm \sqrt{7}$ (No) | - |

| 30 | 29 | $3+5=8 \mid \pm \sqrt{8}$ (No) | - |

| 36 | 35 | $3+6=9 \mid \pm 3 \mid (35, 3), (35, -3)$ |

• **Step 4: Conclusion**

The valid integer pairs (r, k) are:

1. $(5, 2)$
2. $(5, -2)$
3. $(35, 3)$
4. $(35, -3)$

Number of elements in set $S = 4$.

Question: Let $f(x) = \begin{cases} \left| \frac{\sin x}{x} \right| & , x \neq 0 \\ 1 & , x = 0 \end{cases}$. Then total number of critical point(s) in $x \in (-2\pi, 2\pi)$ is

Solution:

• **Step 1: Check $x = 0$**

We know $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. The function is continuous at $x = 0$. However, looking at the derivative of $\frac{\sin x}{x}$, it behaves smoothly at the origin, so $x = 0$ is not a critical point in the sense of non-differentiability here.

• **Step 2: Points where $f(x) = 0$ (Sharp Corners)**

The absolute value $|\cdot|$ creates "sharp corners" (non-differentiable points) whenever the inside expression is zero.

$$\sin x = 0 \implies x = \pm\pi \text{ (within the interval } -2\pi, 2\pi)$$

These are **2 critical points**.

• **Step 3: Points where $f'(x) = 0$ (Local Maxima/Minima)**

For $x \neq 0$ and $x \neq \pm\pi$, we differentiate $g(x) = \frac{\sin x}{x}$:

• **Step 3: Points where $f'(x) = 0$ (Local Maxima/Minima)**

For $x \neq 0$ and $x \neq \pm\pi$, we differentiate $g(x) = \frac{\sin x}{x}$:

$$g'(x) = \frac{x \cos x - \sin x}{x^2} = 0 \implies x = \tan x$$

The equation $x = \tan x$ has solutions near $\pm 1.5\pi$ in our interval.

- One solution in $(\pi, 2\pi)$
- One solution in $(-2\pi, -\pi)$

These are **2 more critical points**.

Total Critical Points: $2 + 2 = 4$

Question: Consider two A.P.s such that A.p.1 has 1st term 1, common difference 5 and total terms 101 and A.P.2 has 1st term 9, common difference 7 and total terms 71 then the number of common terms which appear in both these A.P.s is

Solution:

Step 1: Write the general terms of both APs

- **AP 1:** $a_1 = 1, d_1 = 5, n_1 = 101$
Last term (l_1) = $1 + (101 - 1)5 = 1 + 500 = 501$
- **AP 2:** $a_2 = 9, d_2 = 7, n_2 = 71$
Last term (l_2) = $9 + (71 - 1)7 = 9 + 490 = 499$

The common terms must be less than or equal to the smaller of the two last terms: **Common Terms ≤ 499** .

Step 2: Find the first common term (a_{common})

List the terms of both sequences until a match is found:

- **AP 1:** 1, 6, 11, 16, 21, 26, 31, 36, ...
- **AP 2:** 9, 16, 23, 30, 37, ...

The first common term is 16.

Step 3: Find the Common Difference (D)

The common terms of two APs also form an AP. Its common difference is the **LCM** of the individual common differences:

$$D = \text{LCM}(d_1, d_2) = \text{LCM}(5, 7) = 35$$

Step 4: Find the number of terms (n)

The sequence of common terms is: 16, 51, 86, ...

We use the formula $a + (n - 1)D \leq \text{Max Limit}$:

$$16 + (n - 1)35 \leq 499$$

$$(n - 1)35 \leq 483$$

$$n - 1 \leq \frac{483}{35}$$

$$n - 1 \leq 13.8$$

$$n \leq 14.8$$

Since n must be an integer, the number of common terms is 14.

Question: The domain of $f(x) = \sqrt{\log_{0.6} \left(\frac{|2x - 5|}{\sqrt{|x^2 - 4|}} \right)}$ is

Solution:

• **Step 1: The Inequality**

For the square root and log (base $0.6 < 1$) to be defined:

$$0 < \frac{|2x - 5|}{\sqrt{|x^2 - 4|}} \leq 1$$

This implies: $|2x - 5| \leq \sqrt{|x^2 - 4|}$ and $x \neq \frac{5}{2}, x \neq \pm 2$.

• **Step 2: Squaring and Solving**

$$(2x - 5)^2 \leq |x^2 - 4|$$

This implies: $|2x - 5| \leq \sqrt{|x^2 - 4|}$ and $x \neq \frac{5}{2}, x \neq \pm 2$.

• **Step 2: Squaring and Solving**

$$(2x - 5)^2 \leq |x^2 - 4|$$

Since the denominator in your screenshot has $|x^2 - 4|$, we must consider the case where $x^2 - 4$ is positive ($x > 2$ or $x < -2$):

$$4x^2 - 20x + 25 \leq x^2 - 4$$

$$3x^2 - 20x + 29 \leq 0$$

Using the quadratic formula for $3x^2 - 20x + 29 = 0$:

Using the quadratic formula for $3x^2 - 20x + 29 = 0$:

$$x = \frac{20 \pm \sqrt{400 - 348}}{6} = \frac{20 \pm \sqrt{52}}{6} = \frac{20 \pm 2\sqrt{13}}{6} = \frac{10 \pm \sqrt{13}}{3}$$

• **Step 3: Defining the Interval**

The values are approx 2.13 and 4.53. Both are in the region where $x^2 - 4 > 0$.

Excluding $x = \frac{5}{2}$ (where the log argument becomes 0), we get:

Domain: $\left[\frac{10-\sqrt{13}}{3}, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \frac{10+\sqrt{13}}{3}\right]$

Question: Find midpoint of sides of triangle ABC to $\left(\frac{5}{7}, 7\right), \left(\frac{5}{2}, 3\right)$ and $(4, 5)$. If its incenter is (h, k) then $3h + k$ is equal to

Solution:

• **Step 1: Properties of Midpoint Triangles**

The vertices of $\triangle ABC$ can be found, but a key property is that the **Incenter of the midpoint triangle is NOT the same as the original**, but the sides are parallel and half the length.

• **Step 2: Find Vertices A, B, C**

Using $x_A + x_B = 2x_D$, etc., we find vertices:

$$A = (x_E + x_F - x_D, y_E + y_F - y_D) \dots$$

Alternatively, calculate lengths of the midpoint triangle DEF .

$$DE = \sqrt{\left(\frac{5}{2} - \frac{5}{7}\right)^2 + (3 - 7)^2}, \text{ etc.}$$

- **Note:** This question typically leads to an **Equilateral** or **Right-angled** triangle in JEE to simplify incenter calculation. If DEF is right-angled at F , the incenter formula $\frac{ax_1 + bx_2 + cx_3}{a+b+c}$ is used.

Question: $\lim_{x \rightarrow 2} \frac{\sin(x^3 - 5x^2 + ax + b)}{(\sqrt{x-1} - 1)\log_e(x-1)} = m$, then $a + b + m$ is equal to

Solution:

Problem: $\lim_{x \rightarrow 2} \frac{\sin(8 - 20 + 2a + b)}{(\sqrt{x-1}-1) \log_e(x-1)} = m$. Find $a + b + m$.

- **Step 1: Check the denominator.**

As $x \rightarrow 2$, $(\sqrt{x-1}-1) \rightarrow 0$ and $\log_e(x-1) \rightarrow 0$. For a finite limit m , the numerator must also be 0 at $x = 2$.

$$\sin(8 - 20 + 2a + b) = 0 \implies 2a + b = 12.$$

- **Step 2: Simplify the limit using standard forms.**

Let $x - 2 = h$. As $h \rightarrow 0$:

$$\text{Denominator} \approx (\sqrt{1+h}-1) \log_e(1+h) \approx \left(\frac{h}{2}\right)(h) = \frac{h^2}{2}.$$

Numerator $\approx \sin(P(x))$, where $P(x) = (x-2)^2(x-1)$ because it must have $(x-2)^2$ to cancel the h^2 in the denominator.

$$\text{Expanding } (x-2)^2(x-1) = x^3 - 5x^2 + 8x - 4.$$

Comparing coefficients: $a = 8$ and $b = -4$.

- **Step 3: Calculate m .**

$$\lim_{h \rightarrow 0} \frac{h^2(1+h)}{h^2/2} = 2, \text{ so } m = 2.$$

- **Final Value:** $a + b + m = 8 - 4 + 2 = 6$.

Question: Let $\sum_{k=1}^n a_k = 6n^3$, then $\sum_{k=1}^6 \left(\frac{a_{k+1} - a_k}{36} \right)^2$ equal to

Solution:

- **Step 1:** $a_n = S_n - S_{n-1} = 6n^3 - 6(n-1)^3 = 18n^2 - 18n + 6$.
- **Step 2:** $a_{k+1} - a_k = 18(k+1)^2 - 18(k+1) + 6 - (18k^2 - 18k + 6) = 36k$.
- **Step 3:** $\sum_{k=1}^6 \left(\frac{36k}{36} \right)^2 = \sum_{k=1}^6 k^2 = \frac{6(7)(13)}{6}$.
- **Result:** 91.